

M.TECH MINI-PROJECT REPORT

Project Title:

DESIGN AND FABRICATION OF A SCARA TRAJECTORY MANIPULATOR

Group Members:

- | | |
|------------------------|-----------------|
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1. Executive Summary

This report outlines the project plan for the design and fabrication of a SCARA (Selective Compliance Assembly Robot Arm) trajectory manipulator. The project is aimed at demonstrating foundational skills in mechanical design, mechatronics, and project management, with a specific focus on achieving precise horizontal movement and motion control. The project scope is defined by the objective to create a working prototype using 3D printing within a budget of ₹10,000 and an 8-week timeline. This document details the project's objectives, system architecture, resource allocation, and a comprehensive risk management plan to ensure a successful and timely completion.

2. Project Objectives

The primary objectives of this mini-project are as follows:

- To design a SCARA robotic manipulator capable of precise trajectory-based pick-and-place operations.
- To fabricate the mechanical structure using 3D printing with cost-effective and readily available filament.
- To integrate a control system using a microcontroller and servo motors capable of coordinating SCARA motion.
- To develop and validate a kinematic model of the robotic arm to enable precise trajectory control.
- To successfully manage the project within the stipulated budget of ₹10,000 and the 8-week timeline.

3. Project Parameters

3.1 Scope:

- **Mechanical Design:** The project includes the CAD design of the SCARA arm's links, joints, and end-effector.
- **Fabrication:** The fabrication will be limited to a functional prototype using 3D printing, prioritizing performance over aesthetics.

- **Control System:** The control system will be a microcontroller-based solution, allowing for coordinated two-axis movement and execution of simple pre-programmed trajectories.
- **Kinematics:** A forward and inverse kinematics model will be developed and validated to control the SCARA arm's end-effector position.

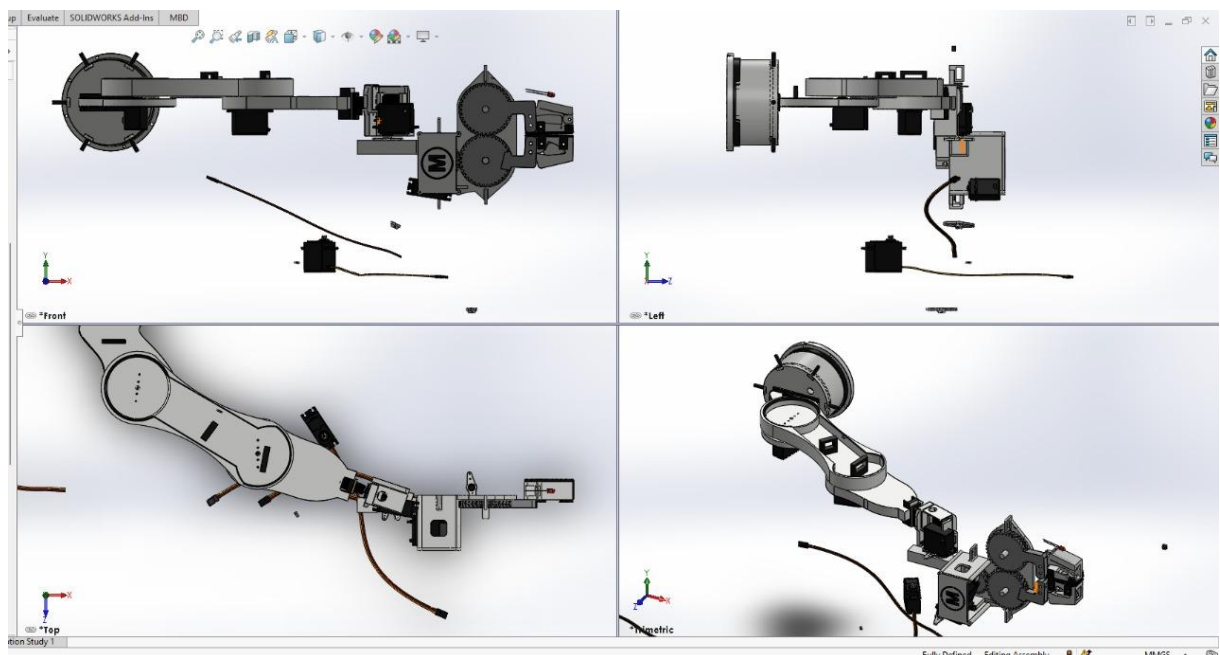
3.2 Constraints:

- **Budget:** The total project budget is strictly limited to ₹10,000.
- **Timeline:** The project must be completed within an 8-week period.
- **Materials:** Material selection is constrained to what is available from local vendors or online retailers within the budget, with a focus on 3D printing filament.
- **Functionality:** The arm's functionality will be limited to pick-and-place operations within a horizontal plane with light-weight objects.

4. System Architecture

The system is composed of three main components: the mechanical subsystem, the electronic control subsystem, and the software subsystem.

- **Mechanical Subsystem:** Consists of the base, links, joints, and gripper. All components are to be fabricated from 3D-printed parts.
- **Electronic Control Subsystem:** An Arduino Uno/Mega will serve as the central processing unit, connected to servo motors via a servo shield or direct wiring. A power supply will be used to drive the motors.
- **Software Subsystem:** The control logic will be implemented in C++ or a similar language using the Arduino IDE. It will include code for commanding coordinated motor movements based on the inverse kinematics model.



5. Hardware Bill of Materials (BOM) & Cost Analysis

Total Budget: ₹10,000

Component	Estimated Quantity	Estimated Unit Cost (₹)	Total Cost (₹)	
Arduino Mega	1	800	800	
High-Torque Servo Motors (MG996R)	5	500	2500	
Standard Servo Motor (SG90) for Gripper	1	150	150	
3D Printing Filament (PLA)	2kg	1000	2000	
Screws, Nuts, and Fasteners	1	500	500	
Wires and Connectors	1	300	300	
Servo Driver Shield	1	400	400	
Power Supply (5V, 5A)	1	600	600	
Miscellaneous (tools, etc.)	-	1500	1500	
Subtotal			₹8,750	

Total Estimated Cost: ₹8,750

6. Procurement Priorities

- **Priority 1 (P1):** Core components essential for functionality. These will be procured in Week 1 to avoid project delays.
- **Priority 2 (P2):** Supporting components required for assembly and integration. These will be purchased once the mechanical design is finalized, around Week 2-3.
- **Priority 3 (P3):** Miscellaneous items and a buffer for unforeseen expenses. These will be procured as needed.

7. Simulation and Validation

- **Design Validation:** The mechanical design will be simulated in SolidWorks to test for joint interference, range of motion, and structural integrity under load.
- **Kinematic Validation:** Both forward and inverse kinematics models will be developed and tested. The calculated end-effector positions will be compared with the simulation results from SolidWorks.
- **Functional Testing:** Once fabricated, the SCARA arm will be tested for its ability to perform coordinated, multi-joint movements and execute specific trajectories for pick-and-place tasks.

8. Project Timeline

Week	Task	Responsible Person(s)
1	Project Planning & Literature Review	
2	Finalize CAD Design & Component Selection	
3	Procurement of P1 Components	
4	3D Printing of Mechanical Parts	
5	Assembly & Integration of Electronics	
6	Software Development & Kinematic Modeling	
7	System Testing & Validation	
8	Report Writing & Final Submission	

9. Task Distribution

- Overall Project Lead, Software Development, Kinematics, and Report Finalization.
- Mechanical Design, 3D Printing, and Assembly.
- Component Procurement, Electronic Integration, and Documentation.

10. Risks and Mitigation

Risk	Probability	Impact	Mitigation Strategy
Component delivery delays	Medium	High	Order from multiple vendors; procure core components early.
Component failure	Low	High	Purchase from a reliable vendor; have a contingency fund for replacements.
Design flaws	Medium	Medium	Thoroughly simulate and validate the design in CAD software before printing.
3D printing failures	Medium	Medium	Perform small test prints; have a plan to recalibrate the printer or adjust print settings.
Budget overrun	Low	High	Adhere strictly to the BOM and track all expenses. Use the contingency fund for emergencies.

11. Deliverables

- **Week 3:** Final CAD model of the SCARA arm.
- **Week 5:** Fabricated and assembled mechanical structure.
- **Week 7:** A working prototype of the robotic arm capable of executing a simple trajectory.
- **Week 8:** Final project report, including all documentation, analysis, and results.

12. Approval Request

This document serves as a formal request for approval to proceed with the project on the **DESIGN AND FABRICATION OF A SCARA TRAJECTORY MANIPULATOR** as per the plan outlined above.

Student Signature : _____

Supervisor Signature : _____